Declaration on the Use of Artificial Intelligence (AI) Tools

Project: Analysis of Library Management System Transactions
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1. Tools and Models Used

- **Google Gemini & ChatGPT:** Used for linguistic and structural support in the written report, and assistance with R code, particularly for details of visualization.
- **DeepL / Google Translate:** Used for translating domain-specific terminology and dataset labels.

2. Nature and Scope of Use

AI tools were exclusively used as supportive tools and not for the core analysis or interpretation of results.

- **Linguistic Revision:** AI tools were applied to find grammar errors, spelling issues, and refine academically concise phrasing in the report, including the Executive Summary and presentation materials. These were used to enhance clarity and conciseness.

- **Structuring:** Assistance was used to improve logical structure and flow of selected sections of the report, particularly in the methodological descriptions and data processing explanations.

- **Code Support and Visualization:** AI tools were used to assist in debugging, syntax correction, and improving code readability in R (e.g., dplyr, ggplot2). In most of cases, support was used for implementing more complex visualizations, such as map-based plots, as well as for graphical elements including layout, font adjustments, and overall design.

3. Verification and Responsibility

All AI-generated suggestions, both in text and code, were critically reviewed, revised, and adapted to only specific context of this project. No content was directly adopted without modification or understanding. The statistical analysis, interpretation of results, and core methodological decisions were carried out independently by the authors. All code was manually tested on the dataset to ensure correctness and consistency of the results. For academic guidelines, the authors bear full responsibility for the entirety of the submitted work and confirm that they fully understand and can explain all parts of the analysis and implementation.